IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Source-case events, mode timeline, and present limitation

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Abstract

The EECON49 file supplies case data and events only; switching remains the project AGSI++. Grid-reference availability is added as a seventh term and all weights become 1/7. All four local supervisors commit GFL→GFM at 20.500 s after the SG trip. The restoration case is fail-closed at 147.020 s, so the reduced model does not yet support a successful reclose/recovery claim. Plots are raw, without noise, smoothing, or clipping.

1 Case and switching rule

The 100-MVA, 60-Hz IEEE 14-bus case has the SG at bus 1 and IBRs at buses 2, 3, 6, and 8. Source-case values are $H = 2.5$ s and $D = 1.0$ pu. The project index is

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|\mathrm{d}f/\mathrm{d}t|}{1.0}, \quad J_P = \frac{|P_{ref} - P|}{0.20},$$

$$J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad J_{GRA} = 1 - GRA, \qquad \mathrm{AGSI}^{++} = \frac{1}{7}\sum J_k.$$

$GRA = 1$ when the SG or a committed GFM reference is available. It is a weighted input, not a hard override. Each IBR uses local hysteresis/dwell: $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$, $T_{d,on} = 0.5$ s, $T_{d,off} = 1.0$ s. The per-device state machine is

$$\mathrm{GFL} \xrightarrow[T_{d,on}]{\mathrm{AGSI}_i^{++} \geq \Gamma_{on}} \mathrm{GFM}, \qquad \mathrm{GFM} \xrightarrow[T_{d,off}]{\mathrm{AGSI}_i^{++} < \Gamma_{off},\ GRA=1} \mathrm{GFL}.$$

Restoration makes the SG/line/load reference available but does not override this state machine.

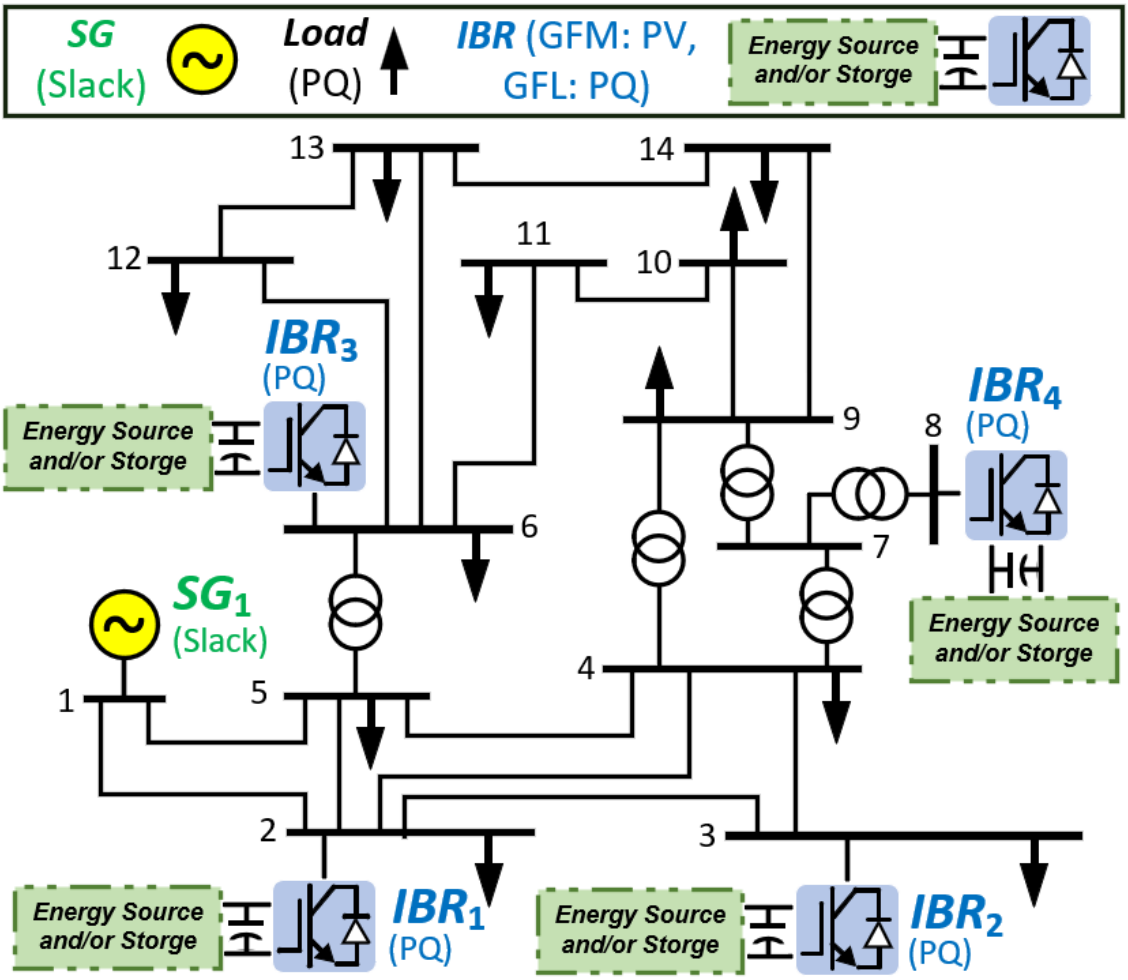

Figure 1: IEEE 14-bus one-line used by the simulation.

2 Power-flow operating point

Table 1: Bus power-flow results (pu)

Bus	Type	$\|V\|$	θ (deg)	P_g	Q_g	P_L	Q_L
1	REF	1.06000	+0.0000	+2.32393	-0.16549	0.00000	0.00000
2	PV	1.04500	-4.9826	+0.40000	+0.43557	0.21700	0.12700
3	PV	1.01000	-12.7251	+0.00000	+0.25075	0.94200	0.19000
4	PQ	1.01767	-10.3129	+0.00000	+0.00000	0.47800	-0.03900
5	PQ	1.01951	-8.7739	+0.00000	+0.00000	0.07600	0.01600
6	PV	1.07000	-14.2209	-0.00000	+0.12731	0.11200	0.07500
7	PQ	1.06152	-13.3596	+0.00000	+0.00000	0.00000	0.00000
8	PV	1.09000	-13.3596	+0.00000	+0.17623	0.00000	0.00000
9	PQ	1.05593	-14.9385	+0.00000	+0.00000	0.29500	0.16600
10	PQ	1.05098	-15.0973	+0.00000	+0.00000	0.09000	0.05800
11	PQ	1.05691	-14.7906	+0.00000	+0.00000	0.03500	0.01800
12	PQ	1.05519	-15.0756	+0.00000	+0.00000	0.06100	0.01600
13	PQ	1.05038	-15.1563	+0.00000	+0.00000	0.13500	0.05800
14	PQ	1.03553	-16.0336	+0.00000	+0.00000	0.14900	0.05000

Table 2: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	+1.568829	-0.204043	+0.042976	+0.072720
2	1	5	+0.755104	+0.038550	+0.027629	+0.060843
3	2	3	+0.732376	+0.035602	+0.023233	+0.051624
4	2	4	+0.561315	-0.015504	+0.016767	+0.014703
5	2	5	+0.415162	+0.011710	+0.009038	-0.009280
6	3	4	-0.232857	+0.044731	+0.003734	-0.003625
7	4	5	-0.611582	+0.158236	+0.005144	+0.016226
8	4	7	+0.280742	-0.096811	-0.000000	+0.017032
9	4	9	+0.160798	-0.004276	-0.000000	+0.013047
10	5	6	+0.440873	+0.124707	+0.000000	+0.044212
11	6	11	+0.073533	+0.035605	+0.000554	+0.001160
12	6	12	+0.077861	+0.025034	+0.000718	+0.001495
13	6	13	+0.177480	+0.072166	+0.002121	+0.004177
14	7	8	-0.000000	-0.171630	+0.000000	+0.004605
15	7	9	+0.280742	+0.057787	+0.000000	+0.008021
16	9	10	+0.052276	+0.042191	+0.000129	+0.000342
17	9	14	+0.094264	+0.036100	+0.001162	+0.002471
18	10	11	-0.037853	-0.016151	+0.000126	+0.000295
19	12	13	+0.016143	+0.007540	+0.000063	+0.000057
20	13	14	+0.056439	+0.017472	+0.000541	+0.001101

3 Event and mode evidence

Time (s)	Event/index/timer	Committed mode
0	normal, timers idle	all GFL
20–20.5	SG trip; each index exceeds Γ_{on} for 0.5 s	all GFL→GFM at 20.500 s
50	all loads +20%	all GFM
85–85.15	three-phase bus-9 fault, $Z_f = 0.01 + j0.01$ pu	all GFM
110	line 6–13 trip	all GFM
145	SG/line restored and load returned to base; no forced handback	remain GFM pending off-dwell
147.020	fail-closed model-validity exit	no validated GFM→GFL

Simultaneous switching is not a broadcast command. The common trip caused all local indices and timers to mature on the same sample. IBR1 is only the plotting-angle reference after the trip; the TDS solves full KCL and no IBR is an algebraic PF slack bus.

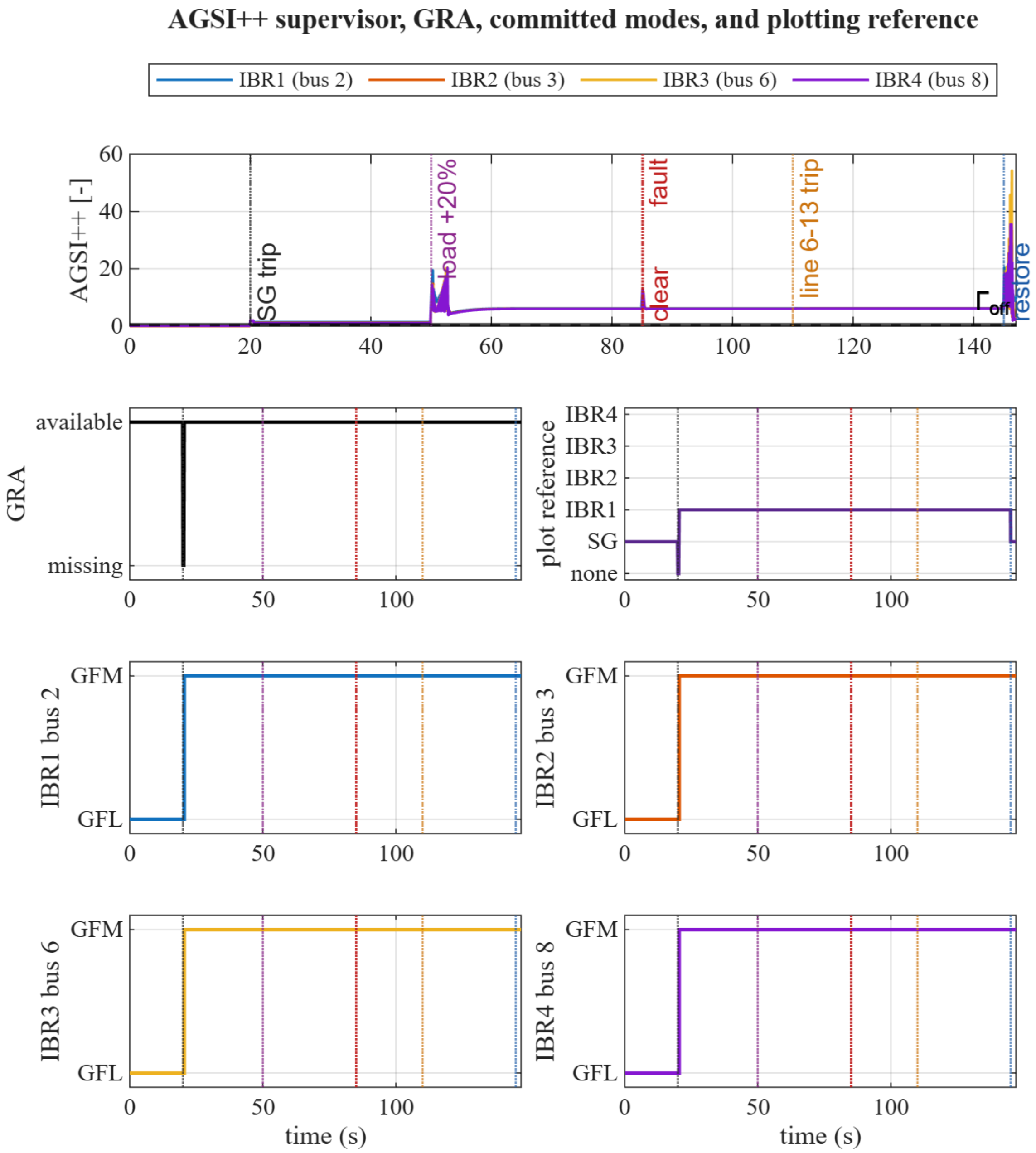

Figure 2: AGSI++, GRA, plotting reference, and separate 0=GFL/1=GFM timelines.

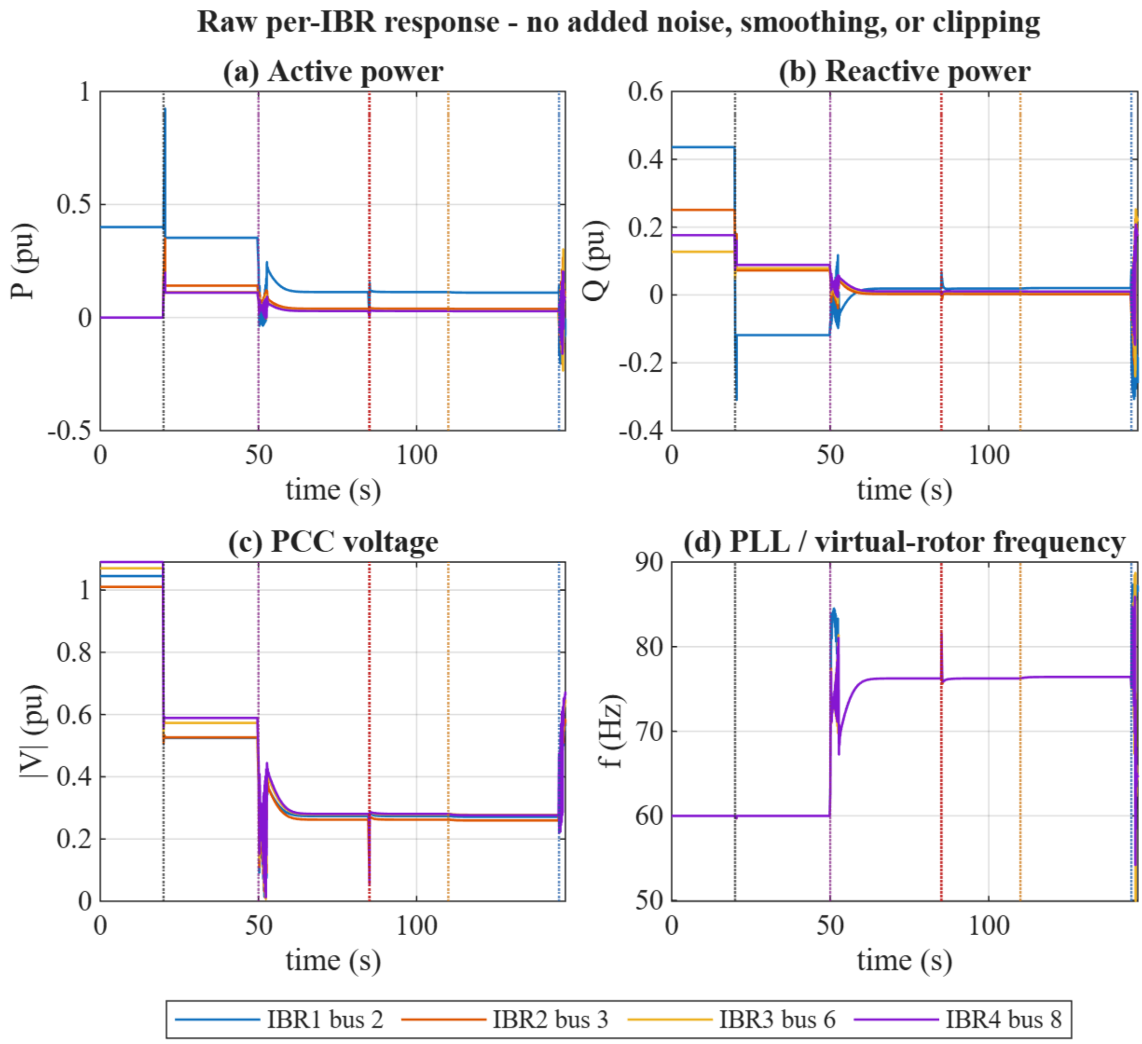

Figure 3: Raw IBR $P, Q, |V|, f$ response ending at the fail-closed point.

4 Present limitation

A physical system should recover after synchronized SG/line restoration and load reset. This study cannot yet demonstrate it: the SG has manual field, no AVR/PSS, no governor/turbine or synchronizer/breaker states, while reduced-6 IBRs omit DC-link and detailed inner/protection dynamics. Reinitializing the SG at the depressed island voltage produces an out-of-domain transient at 147.020 s. The trip-driven GFM entry is validated, but the reclose/recovery claim is withheld. No state reset, parameter tuning, or plot alteration is used to manufacture recovery.

5 Reproduction

```
pf_init_paths;
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures;
```

The generator uses project-owned MATLAB and $\Delta t = 0.005$ s. SSSA and fault-map sections were removed to keep this report focused on IBR switching.